

PROJECT REPORT

ON

Air Quality Analytics: A Comparative and Seasonal Analysis of Pollution Trends Across Indian Cities

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CERTIFICATE

This is to certify that following group of students of have successfully completed the Data Analytics and Visualization project titled Air Quality Analytics: A Comparative and Seasonal Analysis of Pollution Trends Across Indian Cities the requirements of the Bachelors of Technology(B. Tech) program in Artificial Intelligence & Data Science during the academic year 2026-2027.

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GUIDE

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ABSTRACT

Air pollution represents one of the most critical public health and environmental challenges across urban and rural India. Ambient air quality exhibits severe spatio-temporal variability driven by localized emissions, industrialization, meteorological shifts, and seasonal agricultural burning. While regulatory authorities deploy extensive continuous and manual monitoring networks, the resulting data repositories comprise hundreds of thousands of longitudinal observations. Interpreting these high-volume tabular logs via traditional static spreadsheets presents severe bottlenecks in identifying actionable trends, outlier events, and geographical disparities.

This project, titled “**Analysis and Visualization of Air Quality Trends in India Using Power BI,**” details the systematic engineering of a dynamic, interactive Business Intelligence system using day-wise national AQI records sourced from the Central Pollution Control Board (CPCB) via data.gov.in. Spanning 425,279 distinct observational records from 1 May 2015 to 19 June 2025 across 32 States/UTs and 295 monitoring areas, the dataset was processed through a structured Extract, Transform, Load (ETL) pipeline within Power Query. Calculated dimensions and robust Data Analysis Expressions (DAX) were engineered to enable multi-scale dynamic aggregations.

The analytical interface is realized as a unified, four-page dashboard covering:

1. National macro-level metrics,
2. Longitudinal daily and seasonal trends,
3. Multi-tier geographic distributions, and
4. Prominent pollutant group dynamics using decomposition trees and scatter distributions.

The resulting platform translates massive environmental tables into an exploratory decision-support interface for researchers, policymakers, and public stakeholders.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AQI	Air Quality Index
CPCB	Central Pollution Control Board
DAX	Data Analysis Expressions
ETL	Extract, Transform, Load
KPI	Key Performance Indicator
MoEFCC	Ministry of Environment, Forest and Climate Change
NAAQS	National Ambient Air Quality Standards
OGD	Open Government Data Platform India
PM2.5 / PM10	Particulate Matter ($\leq 2.5 \mu\text{m}$ / $\leq 10 \mu\text{m}$)
UI/UX	User Interface / User Experience
UT	Union Territory

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CHAPTER I INTRODUCTION

1.1 General

The Air Quality Index (AQI) is an internationally standardized, unitless numerical metric developed to communicate ambient air quality status and associated short-term health impacts to the general public. In India, the National Air Quality Index framework was officially rolled out under the Swachh Bharat Abhiyan by the Central Pollution Control Board (CPCB) and the Ministry of Environment, Forest and Climate Change (MoEFCC). The Indian AQI system evaluates eight major criteria pollutants: Particulate Matter (PM₁₀, PM_{2.5}), Nitrogen Dioxide (NO₂), Sulfur Dioxide (SO₂), Carbon Monoxide (CO), Ozone (O₃), Ammonia (NH₃), and Lead (Pb).

1.2 Problem Identification

Governmental open-data portals publish expansive, highly authentic longitudinal air quality archives. However, standard access formats present acute operational obstacles:

- **High Data Volume & Tabular Overhead:** The target dataset comprises **425,279 rows**. Conventional spreadsheet utilities (e.g., Microsoft Excel, basic CSV readers) struggle with memory consumption, visual rendering limits, and sluggish formula evaluations over such scale.
- **Absence of Unified Spatio-Temporal Interactivity:** Traditional annual environmental reports deliver pre-aggregated, static PDF tables. These lack the facility to slice multi-year data dynamically, compare specific micro-regions (such as industrial vs. coastal belts), or isolate severe pollution seasons simultaneously.
- **Opaque Pollutant-Category Interrelationships:** Tabular logs list prominent pollutant clusters alongside index values, but fail to visually display the statistical distribution, frequency of occurrence, and hierarchical contributions of specific pollutants to degraded AQI categories.

1.3 Aims & Objectives

The central objective of this research is to construct an enterprise-grade, interactive Business Intelligence dashboard in Microsoft Power BI to extract, clean, model, and visualize air quality dynamics across India between 2015 and 2025.

Specific technical sub-objectives include:

- Performing automated ETL transformations on a 425,279-row national recordset to resolve data types, temporal structures, and missing values.
- Engineering a clean data model leveraging optimized DAX measures for dynamic mathematical aggregation across variable filter contexts.
- Developing a chronologically sorted temporal framework to study longitudinal trends, seasonal winter peaks, and intra-year patterns.
- Implementing spatial visual layers to compare disparities across 32 States/UTs and 295 monitored areas.
- Integrating advanced analytical visuals—including Decomposition Trees, Scatter Plots, and Treemaps—to investigate associations between prominent pollutant groups and AQI severity bands.
- Ensuring zero-latency dynamic cross-filtering via slicers and bidirectional cross-highlighting across four specialized report canvas pages.

1.4 Scope of Project

The scope of this project encompasses:

- Temporal Coverage: Multi-year retrospective analysis spanning 1 May 2015 to 19 June 2025.
- Geographical Span: Standardized records mapped across 32 Indian States/Union Territories encompassing 295 discrete urban and industrial monitoring locations.
- Descriptive and Diagnostic Visual Analytics: Focuses strictly on historical pattern recognition, comparative analysis, spatial variations, and pollutant association tracking.

Explicit Technical Boundaries: The project does not attempt predictive machine learning forecasting, does not possess continuous physical mass concentration logs (e.g., $\mu\text{g}/\text{m}^3$) for individual chemical species, does not deduce medical or epidemiological causality, and is naturally constrained by the physical sensor placement density established by the CPCB across the subcontinent.

CHAPTER II: LITERATURE REVIEW

2.1 Existing System

The prevailing public mechanism for disseminating air quality observations relies primarily on two methodologies:

1. **Static Government Bulletins and Annual Audits:** CPCB, state pollution control boards, and affiliated research institutes regularly publish monthly summaries, technical bulletins, and exhaustive annual reports in PDF format. While methodologically rigorous, these documents offer flat, non-interactive cross-sections. Cross-referencing an area's multi-year seasonal shift against a neighboring state's trajectory requires manual extraction and compilation.
2. **Tabular Open-Data Exports:** Platforms such as data.gov.in provide downloadable flat-file outputs (CSV, XLS). While valuable for raw computational research, these repositories require advanced scripting skills (Python, R) or spreadsheet wrangling to derive high-level visual summaries. For non-programming environmental officers, urban planners, and the public, raw tables act as a barrier to rapid insight discovery.

2.2 Need for Dedicated System

To resolve the bottlenecks of the existing systems, an interactive, zero-code-query analytical reporting system is essential.

A specialized Business Intelligence deployment provides:

- **In-Memory Analytical Engine:** Power BI's VertiPaq engine compresses large datasets into RAM, enabling sub-second multi-dimensional calculations across nearly half a million rows.
- **Cohesive Relational Slicing:** A single interface allows seamless filtering by calendar period, geographical boundary, or health severity tier, with every chart on the canvas responding simultaneously.
- **Cognitive Accessibility:** Visual cues—such as heatmaps, spatial maps, and decomposition trees—allow users to immediately detect anomalies, seasonal winter inversions, and high-pollution regional corridors without parsing tabular text .

CHAPTER III: SYSTEM DESIGN AND METHODOLOGY

3.1 Proposed System Architecture

The system follows a three-stage pipeline: Data Ingestion & Cleaning via Power Query, Data Modeling & DAX Calculation in the VertiPaq engine, and Interactive Visualization across a four-page dashboard layout.

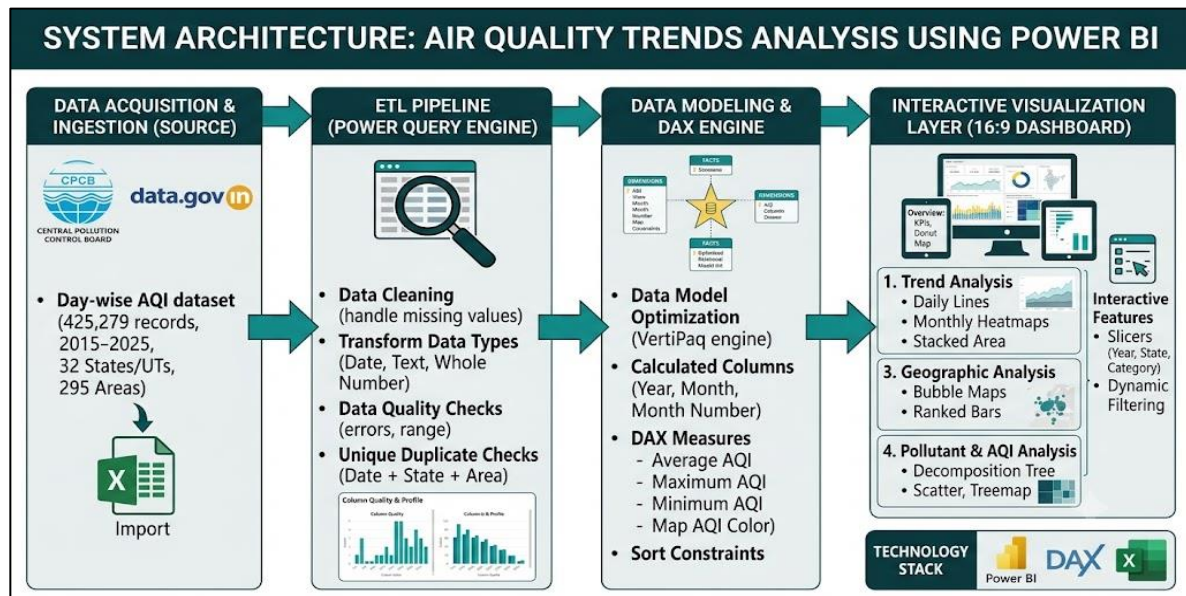


Fig. 3.1 End-to-End System Architecture and Data Pipeline Workflow

3.2 Technology Stack

- **Analytics & Dashboard Engine:** Microsoft Power BI Desktop (x64)
- **ETL & Data Transformation:** Power Query (M Language)
- **Calculations:** Data Analysis Expressions (DAX)
- **Data Ingestion Source:** CPCB via Open Government Data Platform (data.gov.in)

3.3 Module Description

- **Module 1: Data Ingestion & ETL (Power Query):** Ingests the 425,279 raw records, enforces strict schema types (dates, whole numbers, text), manages null values, and validates unique composite keys across dates, states, and monitoring areas.
- **Module 2: Analytical Modeling & DAX Measures:** Constructs the tabular data model and formulates dynamic DAX aggregation metrics—including Average AQI, Maximum AQI, and Minimum AQI—alongside chronological month-sorting logic.
- **Module 3: Visual Analytics Canvas:** Develops visual layouts across the four dashboard pages, incorporating KPI summary cards, multi-year and cyclical line charts, stacked area charts, ranked horizontal bars, maps, treemaps, and decomposition trees.
- **Module 4: Interaction & Navigation:** Configures global synchronized slicers (Year, State/UT, Air Quality Category) and page-navigation elements to support seamless cross-filtering and drill-down exploration.

3.4 Project Work Plan

The project was executed across five sequential phases:

1. Requirement Analysis: Formulating the problem statement, analytical scope, user accessibility criteria, and core research objectives.
2. Data Collection: Sourcing the 10-year historical day-wise AQI dataset from the Central Pollution Control Board (CPCB) via data.gov.in.
3. Data Preprocessing: Transforming raw extracts in Power Query, casting schema data types, validating numerical ranges, and verifying record uniqueness.
4. Dashboard Development: Designing the dimensional structure, authoring DAX calculations, and building visual elements across the four dedicated report canvases.
5. Testing and Validation: Verifying DAX measure accuracy, stress-testing interactive cross-filtering responsiveness, auditing navigation paths, and finalizing the project report.

3.5 Requirement Analysis And Survey Findings

Functional Requirements

- Ingest tabular day-wise records across 10 years without data loss.
- Provide dynamic multi-parameter filtering across temporal, geographic, and categorical boundaries.
- Ensure accurate sorting of calendar months and numerical AQI rankings.

Non-Functional Requirements

- **Query Performance:** Slicer interactions and cross-visual filtering must render within sub-second intervals.
- **Visual Ergonomics:** Standardized color encoding representing recognized CPCB danger levels.
- **Maintainability:** Modular DAX measures and clean ETL query steps enabling continuous refresh cycles.

Survey Findings & Baseline Statistics

Features	Dataset Characteristics
Total Record	425,279 rows
Attributes	8 columns
Observation Window	1 May 2015 to 19 June 2025 (3,690 days)
Spatial Coverage	32 States/UTs; 295 Unique Areas
Global Index Span	Minimum = 3, Maximum = 500, Mean = ~118.66
AQI Categories	Good, Satisfactory, Moderate, Poor, Very Poor, Severe

CHAPTER IV: RESULTS AND DISCUSSION

4.1 Implementation Details

Data transformations were authored in the Power Query M engine. Null and outlier profiling confirmed complete numerical integrity for all 425,279 records in `aqi_value`, while `number_of_monitoring_stations` exhibited occasional missing values that did not impede aggregate calculations. The visual palette was mapped directly to national CPCB reporting standards: Good (Green), Satisfactory (Yellow), Moderate (Orange), Poor (Red), Very Poor (Purple), and Severe (Dark Red).

4.2 System Screenshots

The system includes:

Figure 4.1: Power BI Page 1 — National Air Quality Overview Dashboard. Displays macro KPIs (Average AQI: 118.66, Max: 500, Min: 3, 32 States/UTs, 295 Areas), multi-year trend line (2015–2025), AQI Category Distribution donut chart, Top 10 States/UTs ranking bar chart, and national thematic map.

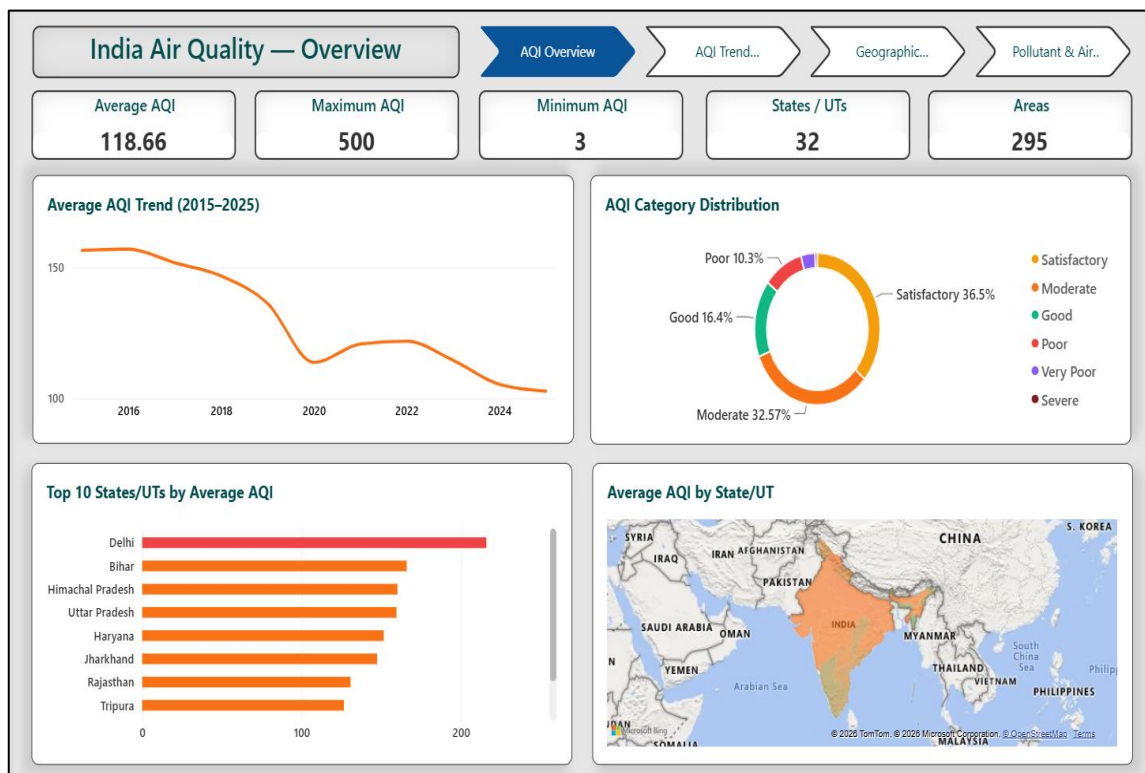


Figure 4.1: Power BI Page 1 — National Air Quality Overview Dashboard.

Figure 4.2: Power BI Page 2 — Multi-Temporal Trend and Heatmap Analysis. Features interactive Year/State/Category slicers, 10-year daily average time-series with a 129.02 baseline reference line, a Year $\times$ Month conditional heatmap matrix, and an AQI category stacked area chart.

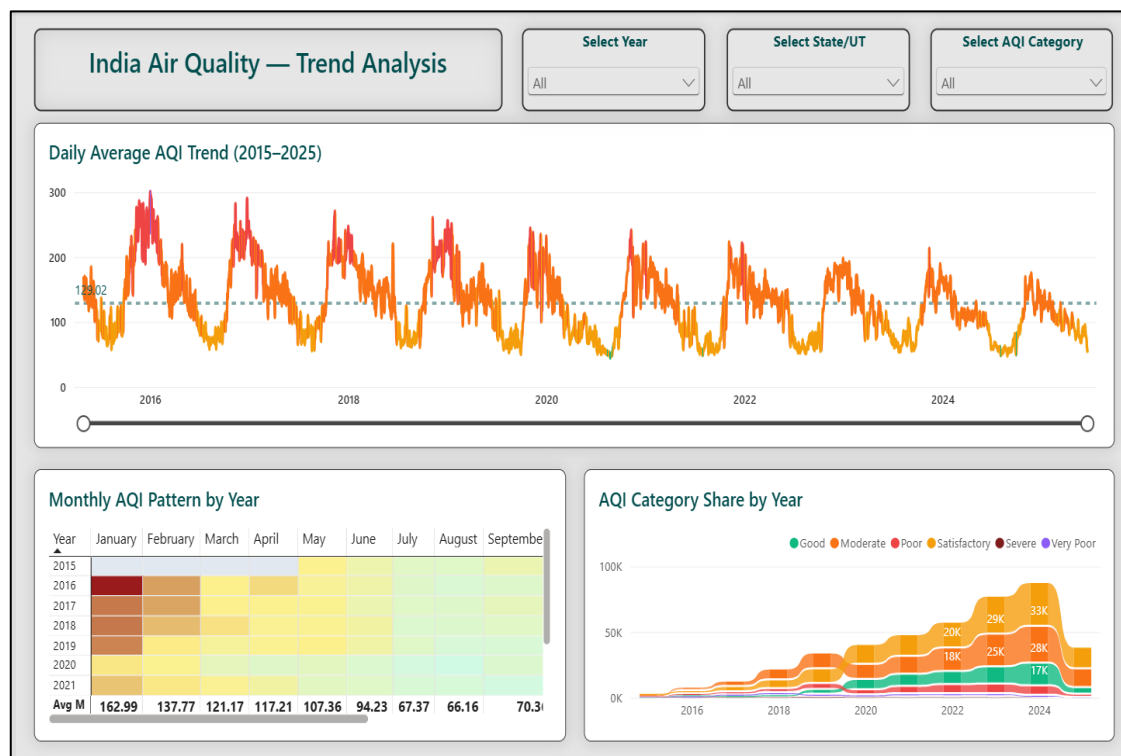


Figure 4.2 : Power BI Page 2 — Multi-Temporal Trend and Heatmap Analysis.

Figure 4.3: Power BI Page 3 — Spatial and Granular Geographic Analysis. Depicts a point map across 295 recorded areas, a detailed State-level metrics matrix, and a ranked horizontal bar chart isolating the top 10 most polluted areas led by Loni Ghaziabad (337) and Jharsuguda (282).

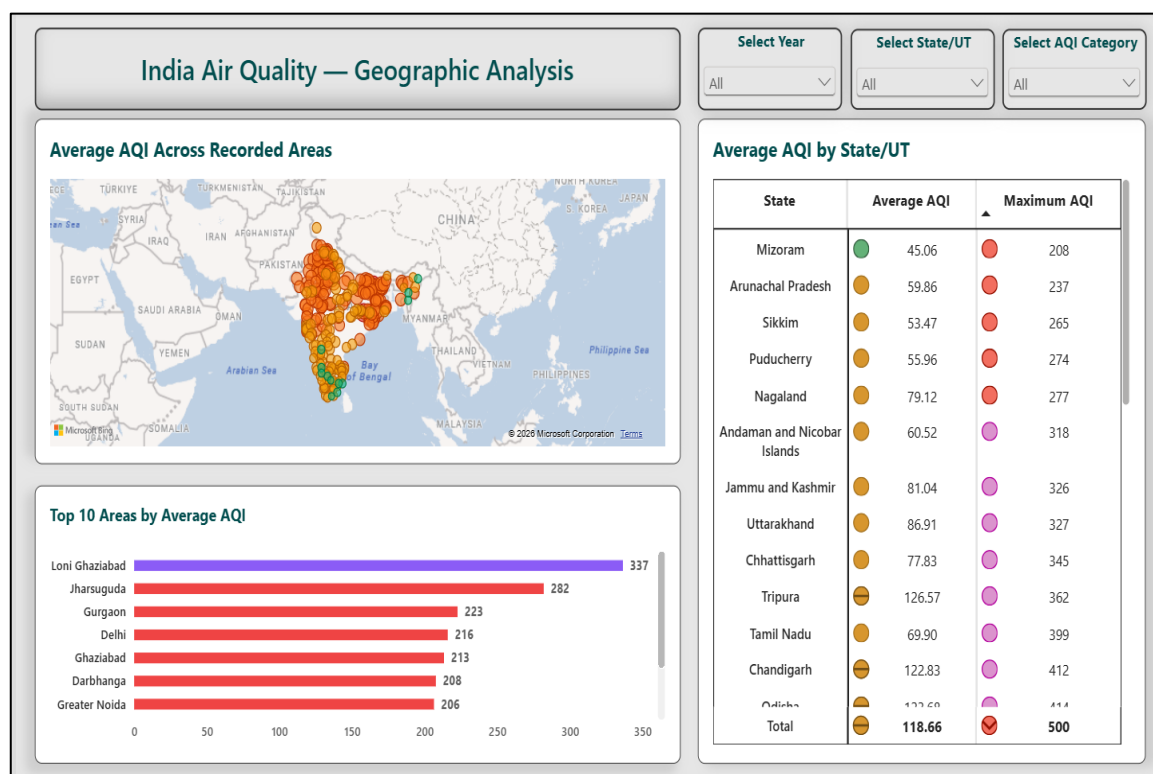


Figure 4.3 : Power BI Page 3 — Spatial and Granular Geographic Analysis

Figure 4.4: Power BI Page 4 — Prominent Pollutant Group and Decomposition Tree Analysis. Features an average AQI by pollutant column chart, an interactive root-cause decomposition tree (drilling into Delhi and particulate matter), a pollutant frequency vs. average severity scatter plot, and a proportional multi-pollutant treemap.

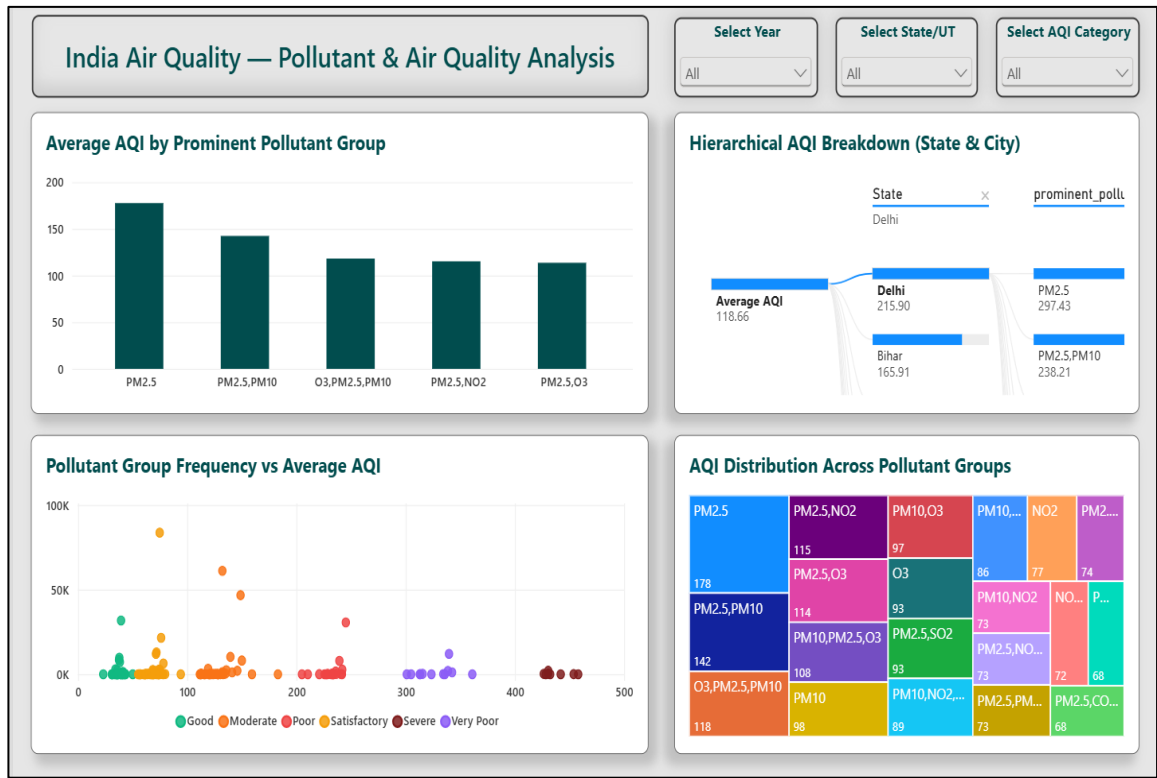


Figure 4.4: Power BI Page 4 — Prominent Pollutant Group and Decomposition Tree Analysis.

4.3 Discussion

The empirical findings from the dashboard reveal several key environmental patterns:

National Baseline vs. Regional Disparities: The overall average AQI of 118.66 places the dataset in the **Moderate** band, but national figures mask sharp geographical divides. Pristine states like Mizoram (45.06) contrast sharply with landlocked northern regions such as Delhi (> 200), which face chronic, severe exposure.

Industrial Hotspots: Granular area monitoring highlights severe localized pollution far above state baselines. Industrial zones like Loni Ghaziabad (337) and Jharsuguda (282) average within the **Very Poor** and upper **Poor** categories due to localized industrial activity, heavy vehicular corridors, and dust.

Seasonal Inversion Cycles: Multi-year timelines show distinct seasonal swings. AQI consistently peaks during winter—averaging 162.99 in January—driven by low boundary layer mixing heights, thermal inversions, and agricultural residue burning, before dropping to annual lows during the monsoon in August (66.16) due to rain precipitation washing out particulates.

Dominance of Fine Particulates: Visual diagnostics confirm that PM_{2.5} is the primary driver of hazardous air quality. Episodes dominated by PM_{2.5} record the highest average index value (178), pushing regional averages to severe levels (297.43 in Delhi), demonstrating that emissions mitigation must focus primarily on fine combustion particles.

CHAPTER V: CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

This project successfully designed and implemented an enterprise Business Intelligence platform using Microsoft Power BI to analyze 425,279 day-wise national AQI observations recorded by the CPCB between May 2015 and June 2025.

By structuring the data through an automated Power Query pipeline and developing a dedicated DAX-driven model, the raw dataset was transformed into an interactive, four-page decision-support interface. The system balances high-level national performance summaries with granular diagnostic tools, including decomposition trees, temporal heatmaps, and spatial maps. The resulting platform demonstrates how modern data visualization workflows can make complex, large-scale environmental open data transparent, accessible, and actionable for both technical and non-technical stakeholders.

5.2 Future Scope

The capabilities developed in this project provide a foundation that can be expanded through several technical enhancements:

- **Live Streaming Pipelines:** Replacing static Excel exports with REST API connectors to the CPCB server or direct Web hooks on data.gov.in, using Power BI Scheduled Refresh to enable near real-time operational monitoring.
- **Granular Species Modeling:** Integrating datasets that contain continuous physical mass concentrations ($\mu\text{g}/\text{m}^3$) for all eight criteria pollutants ($\text{PM}_{2.5}$, PM_{10} , NO_2 , SO_2 , CO , O_3 , NH_3 , Pb), enabling multi-pollutant correlation matrices.
- **Predictive Forecasting Integration:** Training machine learning regression and time-series models (e.g., ARIMA, Prophet, Long Short-Term Memory networks) within Python/R visual scripts to project 7-day forward-looking AQI forecasts directly onto the dashboard canvas.
- **Meteorological Cross-Analysis:** Merging ambient air quality metrics with concurrent meteorological variables—including ambient temperature, relative humidity, planetary boundary layer height, wind speed, and precipitation records—to model and quantify environmental dispersion dynamics.

CHAPTER VI : REFERENCES

- Central Pollution Control Board (CPCB), Ministry of Environment, Forest and Climate Change, Government of India. *National Air Quality Index (AQI) Report and Standards Documentation*. New Delhi, India.
- Open Government Data (OGD) Platform India (data.gov.in), National Informatics Centre (NIC). *Day-wise Air Quality Index Records of Indian Cities/Areas (2015–2025)*.
- Microsoft Corporation. *Power BI Documentation: Model data, optimize DAX formulas, and design accessible reporting dashboards*. Microsoft Learn Technical Repository.
- Marco Russo and Alberto Ferrari. *The Definitive Guide to DAX: Business intelligence with Microsoft Power Pivot, SQL Server Analysis Services Tabular, and Power BI*. 2nd Edition, Microsoft Press.
- World Health Organization (WHO). *WHO Global Air Quality Guidelines: Particulate matter (PM_{2.5} and PM₁₀), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide*. Geneva: World Health Organization.
- Ministry of Environment, Forest and Climate Change (MoEFCC), Government of India. *National Clean Air Programme (NCAP) Operational Guidelines and Framework*. New Delhi, India.